

Measures

Last modified: 2026-10-09 09:50:57 (PDT)

Remark. A measure assigns a size to each set in a [collection of sets](#): how many elements it has, how long it is, or how likely it is. This page builds measures from σ -algebras and additivity, using the [sets and functions](#) page's definitions. One kind of measure, the probability measure (Definition 0.19), is the starting point of the Morrison Lab's probability notes ([probability measure](#)).

σ -algebras

Definition 0.1 (σ -algebra). A σ -algebra on a set S is a [collection](#) $\mathcal{S}$ of [subsets](#) of S that satisfies:

- $\mathcal{S}$ contains S itself.
- For each set A in $\mathcal{S}$, $\mathcal{S}$ contains its [complement](#) $S \setminus A$.
- For each [sequence](#) $A_1, A_2, \dots$ of sets in $\mathcal{S}$, $\mathcal{S}$ contains their union $\bigcup_{i=1}^{\infty} A_i$.

Remark 0.1 (σ -field). Other sources call a σ -algebra a σ -**field** (see [Wikipedia: \$\sigma\$ -algebra](#)). For example, Billingsley (1995) uses “ σ -field”. The two names mean the same thing.

Example 0.1 (σ -algebras for die rolls). For the die rolls $D = \{1, 2, 3, 4, 5, 6\}$ ([sets of die rolls](#)), the collection of all subsets of D is a σ -algebra on D , since complements and unions of subsets of D are again subsets of D . So is the smaller collection

$$\{\emptyset, \{2, 4, 6\}, \{1, 3, 5\}, D\}$$

which contains D , the complement of each of its sets, and every union of its sets: for example, $\{2, 4, 6\} \cup \{1, 3, 5\} = D$.

Example 0.2 (A collection that is not a σ -algebra). The collection $\{\emptyset, \{1\}, D\}$ is not a σ -algebra on $D = \{1, 2, 3, 4, 5, 6\}$: it contains $\{1\}$ but not its complement $D \setminus \{1\} = \{2, 3, 4, 5, 6\}$.

Definition 0.2 (Closed under an operation (closure)). A set A is **closed under** an operation, such as addition of numbers or union of sets, if applying the operation to elements of A always gives an element of A . This property is called **closure** under the operation.

Example 0.3 (Closed and not closed).

- The **even** integers are closed under addition: for example, $2 + 4 = 6$ is even, and $2m + 2n = 2(m + n)$ is even for all integers m and n .
- The odd integers are not closed under addition: 1 and 3 are odd, but $1 + 3 = 4$ is even.
- A **σ -algebra** $\mathcal{S}$ on a set S is closed under complements in S and under unions of sequences of its sets, because Definition 0.1 requires both. The collection in Example 0.2 is not closed under complements.

Theorem 0.1 (Closure properties of a σ -algebra). If $\mathcal{S}$ is a **σ -algebra** on a set S , then $\mathcal{S}$ contains:

- the **empty set** $\emptyset$;
- the union $A_1 \cup \dots \cup A_n$ of any **finitely many** sets $A_1, \dots, A_n$ in $\mathcal{S}$;
- the intersection $\bigcap_{i=1}^{\infty} A_i$ of any sequence $A_1, A_2, \dots$ of sets in $\mathcal{S}$.

So $\mathcal{S}$ is **closed under** finite unions and under intersections of sequences of its sets.

i Proof

Proof. **Empty set.** $\mathcal{S}$ contains S , so it contains the complement of S , which is $S \setminus S = \emptyset$. **Finite unions.** Extend $A_1, \dots, A_n$ to a sequence by setting $A_{n+1} = A_{n+2} = \dots = \emptyset$; each of these sets is in $\mathcal{S}$, by the first part. Then:

$$\begin{aligned} A_1 \cup \dots \cup A_n &= A_1 \cup \dots \cup A_n \cup \emptyset \cup \emptyset \cup \dots \quad (\text{a union with } \emptyset \text{ adds no elements}) \\ &= \bigcup_{i=1}^{\infty} A_i \quad (A_i = \emptyset \text{ for } i > n) \end{aligned}$$

and $\mathcal{S}$ contains $\bigcup_{i=1}^{\infty} A_i$ by the union rule of Definition 0.1.

Countable intersections. Each A_i is a subset of S , so an element of S is outside $S \setminus A_i$

exactly when it is in A_i ; that is, $S \setminus (S \setminus A_i) = A_i$. So, by [De Morgan's laws](#):

$$\begin{aligned} S \setminus \bigcup_{i=1}^{\infty} (S \setminus A_i) &= \bigcap_{i=1}^{\infty} (S \setminus (S \setminus A_i)) \quad (\text{De Morgan's laws}) \\ &= \bigcap_{i=1}^{\infty} A_i \quad (S \setminus (S \setminus A_i) = A_i) \end{aligned}$$

Each $S \setminus A_i$ is in $\mathcal{S}$ by the complement rule, so their union is in $\mathcal{S}$ by the union rule, and the complement of that union is in $\mathcal{S}$ by the complement rule again. $\square$

Theorem 0.2 (All subsets form a σ -algebra). *For any set S , the collection of all subsets of S is a σ -algebra on S .*

Proof

Proof. Each condition of Definition [0.1](#) holds:

- S is a subset of itself.
- For each subset A of S , the complement $S \setminus A$ contains only elements of S , so it is a subset of S .
- For each sequence $A_1, A_2, \dots$ of subsets of S , every element of $\bigcup_{i=1}^{\infty} A_i$ is in some A_i and so in S ; the union is a subset of S .

$\square$

Theorem 0.3 (An intersection of σ -algebras is a σ -algebra). *Let $\mathcal{S}_j$ be a σ -algebra on a set S for each j in a nonempty set of indices J . Then their [intersection](#) $\bigcap_{j \in J} \mathcal{S}_j$, the collection of subsets of S that are in every $\mathcal{S}_j$, is a σ -algebra on S .*

Proof

Proof. Each condition of Definition [0.1](#) holds:

- S is in every $\mathcal{S}_j$, so it is in the intersection.
- If A is in the intersection, then A is in every $\mathcal{S}_j$, so its complement $S \setminus A$ is in every $\mathcal{S}_j$, and so it is in the intersection.
- If $A_1, A_2, \dots$ are in the intersection, then they are in every $\mathcal{S}_j$, so their union $\bigcup_{i=1}^{\infty} A_i$ is in every $\mathcal{S}_j$, and so it is in the intersection.

$\square$

Definition 0.3 (σ -algebra generated by a collection). Let $\mathcal{C}$ be a **collection** of subsets of a set S . The **σ -algebra generated by $\mathcal{C}$** , written $\sigma(\mathcal{C})$, is the intersection of all the σ -algebras on S that contain every set in $\mathcal{C}$.

Remark 0.2 (The smallest σ -algebra containing $\mathcal{C}$). $\sigma(\mathcal{C})$ is a σ -algebra by Theorem 0.3 (the collection of all subsets of S is one of the σ -algebras intersected, by Theorem 0.2, so there is at least one). It contains every set in $\mathcal{C}$, and it is contained in every other σ -algebra on S that does, so it is the smallest σ -algebra on S that contains $\mathcal{C}$.

Example 0.4 (The σ -algebra generated by the even rolls). For the die rolls $D = \{1, 2, 3, 4, 5, 6\}$, let $\mathcal{C} = \{\{2, 4, 6\}\}$. Every σ -algebra on D that contains $\{2, 4, 6\}$ must also contain D , the complement $\{1, 3, 5\}$, and $\emptyset$ (Definition 0.1 and Theorem 0.1). The collection $\{\emptyset, \{2, 4, 6\}, \{1, 3, 5\}, D\}$ is already a σ -algebra (Example 0.1), so it is $\sigma(\mathcal{C})$.

Definition 0.4 (Borel σ -algebra). The **Borel σ -algebra** on $\mathbb{R}$, written $\mathcal{B}$, is the σ -algebra generated by (Definition 0.3) the collection of all **intervals**. Its sets are called **Borel sets**.

Example 0.5 (Some Borel sets).

- Every interval, such as $[0, 1]$, $(2, 5)$, or $[0, \infty)$, is a Borel set.
- A single point $\{a\} = [a, a]$ is an interval, so it is a Borel set.
- The **integers** are a Borel set: they are the union of the sequence of single points $\{0\}, \{1\}, \{-1\}, \{2\}, \{-2\}, \dots$ (**countable and uncountable sets** lists them this way), and $\mathcal{B}$ contains that union by Definition 0.1.
- The complement $\mathbb{R} \setminus [0, 1]$ is a Borel set, because $\mathcal{B}$ contains the complement of each of its sets.

Pairwise disjoint sets

Definition 0.5 (Pairwise disjoint sets (disjoint, mutually disjoint)). Finitely or countably many sets $A_1, A_2, \dots$ are **pairwise disjoint** (also called **disjoint** or **mutually disjoint**) when no two of them share an element:

$$A_i \cap A_j = \emptyset \quad \text{for all } i \neq j$$

Example 0.6 (Low and high die rolls). For the die rolls $D = \{1, 2, 3, 4, 5, 6\}$ (sets of die rolls), the sets $\{1, 2\}$ and $\{5, 6\}$ are pairwise disjoint: no roll is in both. The sets $\{2, 4, 6\}$ and $\{1, 2\}$ are not, because 2 is in both.

Infinite sums

Definition 0.6 (Non-decreasing sequence). A sequence $a_1, a_2, \dots$ of real numbers or of extended non-negative reals is **non-decreasing** if each term is at least as large as the one before it:

$$a_n \leq a_{n+1} \quad \text{for every } n \in \mathbb{N}$$

Example 0.7 (Non-decreasing and not).

- $1, 2, 2, 3, 3, 3, \dots$ is non-decreasing: a term may equal the one before it.
- $1, 2, \infty, \infty, \dots$ is non-decreasing, because ∞ is greater than every real number.
- $1, \frac{1}{2}, \frac{1}{3}, \dots$ is not non-decreasing, because $a_2 = \frac{1}{2} < 1 = a_1$.

Definition 0.7 (Partial sum). Let $a_1, a_2, \dots$ be a sequence of real numbers or of extended non-negative reals. For each $n \in \mathbb{N}$, its n th **partial sum** is the sum of its first n terms:

$$s_n \stackrel{\text{def}}{=} \sum_{i=1}^n a_i = a_1 + \dots + a_n$$

Example 0.8 (Partial sums of halves). For $a_i = 1/2^i$, the sequence $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \dots$, the first three partial sums are $s_1 = \frac{1}{2}$, $s_2 = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$, and $s_3 = \frac{3}{4} + \frac{1}{8} = \frac{7}{8}$. In general, $s_n = 1 - 1/2^n$.

Definition 0.8 (Infinite sum (series)). Let $a_1, a_2, \dots$ be a sequence in $[0, \infty]$, with partial sums s_n (Definition 0.7). The **infinite sum** (also called a **series**) $\sum_{i=1}^{\infty} a_i$ is the limit of the partial sums:

$$\sum_{i=1}^{\infty} a_i \stackrel{\text{def}}{=} \lim_{n \rightarrow \infty} s_n$$

Here the limit is a real number when the partial sums are real numbers that converge to it, and it is ∞ when the partial sums diverge to ∞ or some partial sum is ∞ ; Lemma 0.1, next, shows that one of these always happens. Sums with other index ranges, such as

$\sum_{k=0}^{\infty} a_k$, are defined the same way.

Example 0.9 (A finite and an infinite sum).

- For $a_i = 1/2^i$, the partial sums $s_n = 1 - 1/2^n$ (Example 0.8) converge to 1, because $1/2^n \rightarrow 0$. So $\sum_{i=1}^{\infty} 1/2^i = 1$.
- For $a_i = 1$, the partial sums are $s_n = n$, which diverge to ∞ . So $\sum_{i=1}^{\infty} 1 = \infty$.

Lemma 0.1 (Sums of non-negative terms). *Let $a_1, a_2, \dots$ be values in $[0, \infty]$, with partial sums s_n (Definition 0.7), where $x + \infty = \infty$ for every x in $[0, \infty]$. Then the partial sums are **non-decreasing**, $s_1 \leq s_2 \leq \dots$, so their **limit**, the infinite sum $\sum_{i=1}^{\infty} a_i$ (Definition 0.8), always exists, and it is either a finite number or ∞ .*

Proof

Proof. For each n :

$$\begin{aligned} s_{n+1} &= s_n + a_{n+1} && \text{(definition of } s_{n+1}) \\ &\geq s_n && (a_{n+1} \geq 0) \end{aligned}$$

If some partial sum s_N is ∞ , then $s_n = \infty$ for every $n \geq N$, so the limit is ∞ . Otherwise, the partial sums form a non-decreasing sequence of **real numbers**. If that sequence is **bounded above**, it **converges** to a finite number (see [Wikipedia: Monotone convergence theorem](#)). If it is not bounded above, then for every number M some s_N exceeds M , and so does every later $s_n \geq s_N$; the limit is ∞ . $\square$

Additivity

Definition 0.9 (Finite additivity). Let $\mathcal{S}$ be a **σ -algebra** on a set S . A **function** $\mu : \mathcal{S} \rightarrow [0, \infty]$, with values in the **extended non-negative reals**, is **finitely additive** if, for every finite collection of **pairwise disjoint** sets $A_1, \dots, A_n$ in $\mathcal{S}$, the value of their union is the sum of their values:

$$\mu(A_1 \cup \dots \cup A_n) = \sum_{i=1}^n \mu(A_i)$$

Example 0.10 (Counting elements is finitely additive). For the die rolls $D = \{1, 2, 3, 4, 5, 6\}$ (sets of die rolls), let $\mu(A) \stackrel{\text{def}}{=} |A|$, the number of elements of A (cardinality), for each set A in the σ -algebra of all subsets of D (Example 0.1). The sets $\{1, 2\}$ and $\{5, 6\}$ are pairwise disjoint (Example 0.6), and:

$$\begin{aligned}\mu(\{1, 2\} \cup \{5, 6\}) &= \mu(\{1, 2, 5, 6\}) && \text{(take the union)} \\ &= 4 && \text{(count the elements)} \\ &= 2 + 2 && \text{(write 4 as a sum)} \\ &= \mu(\{1, 2\}) + \mu(\{5, 6\}) && \text{(count each set's elements)}\end{aligned}$$

The same holds for any pairwise disjoint sets, because the sizes of disjoint sets add, so μ is finitely additive.

Example 0.11 (Squaring the count is not finitely additive). On the same σ -algebra, let $\nu(A) \stackrel{\text{def}}{=} |A|^2$. The sets $\{1\}$ and $\{2\}$ are pairwise disjoint, but

$$\begin{aligned}\nu(\{1\} \cup \{2\}) &= \nu(\{1, 2\}) && \text{(take the union)} \\ &= 2^2 \\ &= 4, && \text{(count, then square)} \\ \nu(\{1\}) + \nu(\{2\}) &= 1^2 + 1^2 \\ &= 2, && \text{(count each set, then square)}\end{aligned}$$

and $4 \neq 2$, so ν is not finitely additive.

Definition 0.10 (Countable additivity (σ -additivity)). Let $\mathcal{S}$ be a σ -algebra on a set S . A function $\mu : \mathcal{S} \rightarrow [0, \infty]$ is **countably additive** (also called **σ -additive**) if, for every sequence of pairwise disjoint sets $A_1, A_2, \dots$ in $\mathcal{S}$, the value of their union is the sum of their values:

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i)$$

Example 0.12 (Counting elements is countably additive). For the counting function $\mu(A) \stackrel{\text{def}}{=} |A|$ of Example 0.10, take any sequence of pairwise disjoint sets $A_1, A_2, \dots$. No two of them share an element, and D has only six elements, so at most six of the A_i contain any elements; the rest are $\emptyset$, with $\mu(\emptyset) = 0$. The infinite sum therefore has at most six nonzero terms, and finite additivity (Example 0.10) shows that those terms add up to μ of the union. So μ is countably additive.

Remark 0.3 (The sum in countable additivity always has a value). By Lemma 0.1, the right-hand side of the equation in Definition 0.10, $\sum_{i=1}^{\infty} \mu(A_i)$, always has a value, because each $\mu(A_i)$ is in $[0, \infty]$. For example, if $\mu(A_i) = 1/2^i$ for each i , the partial sums are $s_n = 1 - 1/2^n$, and the sum is 1; if $\mu(A_i) = 1$ for each i , the partial sums are $s_n = n$, and the sum is ∞ .

Lemma 0.2 (Countable additivity and the empty set). *If μ is a countably additive function on a σ -algebra $\mathcal{S}$, then $\mu(\emptyset) = 0$ or $\mu(\emptyset) = \infty$.*

i Proof

Proof. The set $\emptyset = S \setminus S$ is in $\mathcal{S}$, as the complement of S , and the sequence $\emptyset, \emptyset, \dots$ is pairwise disjoint with union $\emptyset$, so:

$$\begin{aligned} \mu(\emptyset) &= \mu\left(\bigcup_{i=1}^{\infty} \emptyset\right) \quad (\text{the union of copies of } \emptyset \text{ is } \emptyset) \\ &= \sum_{i=1}^{\infty} \mu(\emptyset) \quad (\text{countable additivity}) \end{aligned}$$

If $\mu(\emptyset) = c$ for a finite $c > 0$, the right-hand side is $c + c + \dots = \infty \neq c$, a contradiction. So $\mu(\emptyset)$ is 0 or ∞ . □

Example 0.13 (A countably additive function with $\mu(\emptyset) = 0$). The counting function $\mu(A) \stackrel{\text{def}}{=} |A|$ of Example 0.12 is countably additive, and:

$$\begin{aligned} \mu(\emptyset) &= |\emptyset| \quad (\text{definition of } \mu) \\ &= 0 \quad (\emptyset \text{ has no elements}) \end{aligned}$$

Example 0.14 (A countably additive function with $\mu(\emptyset) = \infty$). For the die rolls $D = \{1, 2, 3, 4, 5, 6\}$ (sets of die rolls), let $\mu(A) \stackrel{\text{def}}{=} \infty$ for every set A in the σ -algebra of all subsets of D (Example 0.1), including $A = \emptyset$. For any sequence of pairwise disjoint sets $A_1, A_2, \dots$, the left-hand side of the countable additivity equation is:

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \infty \quad (\text{definition of } \mu)$$

and the right-hand side, the limit of its partial sums (Lemma 0.1), is:

$$\begin{aligned}\sum_{i=1}^{\infty} \mu(A_i) &= \infty + \infty + \dots \quad (\text{definition of } \mu) \\ &= \infty \quad (\text{every partial sum is } \infty)\end{aligned}$$

The two sides agree, so μ is countably additive, and $\mu(\emptyset) = \infty$.

Remark 0.4 ($\mu(\emptyset) = 0$ is an extra requirement). Example 0.13 and Example 0.14 show that both values allowed by Lemma 0.2 occur for countably additive functions. So $\mu(\emptyset) = 0$ is an extra requirement, not a consequence of countable additivity.

Theorem 0.4 (Countable additivity implies finite additivity). *If μ is a countably additive function on a σ -algebra $\mathcal{S}$, and $\mu(\emptyset) = 0$, then μ is finitely additive.*

i Proof

Proof. Let $A_1, \dots, A_n$ be pairwise disjoint sets in $\mathcal{S}$, and extend them to a sequence by setting $A_{n+1} = A_{n+2} = \dots = \emptyset$. The extended sequence is still pairwise disjoint, since $\emptyset$ shares no element with any set, and its union is $A_1 \cup \dots \cup A_n$. So:

$$\begin{aligned}\mu(A_1 \cup \dots \cup A_n) &= \mu\left(\bigcup_{i=1}^{\infty} A_i\right) && (A_i = \emptyset \text{ for } i > n) \\ &= \sum_{i=1}^{\infty} \mu(A_i) && (\text{countable additivity}) \\ &= \sum_{i=1}^n \mu(A_i) + \sum_{i=n+1}^{\infty} \mu(\emptyset) && (A_i = \emptyset \text{ for } i > n) \\ &= \sum_{i=1}^n \mu(A_i) && (\mu(\emptyset) = 0)\end{aligned}$$

□

Example 0.15 (Finitely additive but not countably additive). Let $S = \{0, 1, 2, \dots\}$, with the σ -algebra of all subsets of S (Theorem 0.2), and define:

$$\mu(A) \stackrel{\text{def}}{=} \begin{cases} 0 & \text{if } A \text{ is finite} \\ \infty & \text{if } A \text{ is infinite} \end{cases}$$

μ is finitely additive. Let $A_1, \dots, A_n$ be pairwise disjoint subsets of S . If every A_i is finite, then so is their union, and:

$$\begin{aligned}
\mu(A_1 \cup \dots \cup A_n) &= 0 && \text{(a finite union of finite sets is finite)} \\
&= \sum_{i=1}^n 0 && \text{(a sum of zeros is 0)} \\
&= \sum_{i=1}^n \mu(A_i) && \text{(each } A_i \text{ is finite)}
\end{aligned}$$

If some A_k is infinite, then the union, which contains A_k , is infinite too, and:

$$\begin{aligned}
\mu(A_1 \cup \dots \cup A_n) &= \infty && \text{(the union is infinite)} \\
&= \mu(A_k) + \sum_{i \in \{1, \dots, n\} \setminus \{k\}} \mu(A_i) && (\mu(A_k) = \infty, \text{ and } \infty + x = \infty) \\
&= \sum_{i=1}^n \mu(A_i) && \text{(regroup the terms)}
\end{aligned}$$

μ is not countably additive. The single-element sets $\{0\}, \{1\}, \{2\}, \dots$ are pairwise disjoint, and their union is S , which is infinite. So:

$$\begin{aligned}
\mu\left(\bigcup_{k=0}^{\infty} \{k\}\right) &= \mu(S) && \text{(the union is } S) \\
&= \infty && (S \text{ is infinite})
\end{aligned}$$

but:

$$\begin{aligned}
\sum_{k=0}^{\infty} \mu(\{k\}) &= 0 + 0 + \dots && \text{(each } \{k\} \text{ is finite)} \\
&= 0 && \text{(every partial sum is 0)}
\end{aligned}$$

Remark 0.5 (Finite additivity does not imply countable additivity). Example 0.15 shows that the converse of Theorem 0.4 fails, even when $\mu(\emptyset) = 0$: a finitely additive function need not be countably additive (see [Wikipedia: Sigma-additive set function](#), “An additive function which is not σ -additive”). The function μ in Example 0.15 has $\mu(\emptyset) = 0$, because $\emptyset$ is finite.

Measures

Definition 0.11 (Measure). A **measure** on a set S with a σ -algebra $\mathcal{S}$ is a function $\mu : \mathcal{S} \rightarrow [0, \infty]$ that satisfies:

- $\mu(\emptyset) = 0$.
- μ is **countably additive**.

Example 0.16 (Counting elements is a measure). The counting function $\mu(A) \stackrel{\text{def}}{=} |A|$ of Example 0.10, defined on the σ -algebra of all subsets of D (Example 0.1), takes values in $[0, \infty]$, is countably additive (Example 0.12), and gives $\mu(\emptyset) = 0$, so it is a measure.

Corollary 0.1 (Measures are finitely additive). *Every measure is finitely additive.*

i Proof

Proof. A measure μ is countably additive and has $\mu(\emptyset) = 0$ (Definition 0.11), so Theorem 0.4 applies to it. $\square$

Definition 0.12 (Counting measure). The **counting measure** on a set S is the **measure** on the σ -algebra of all subsets of S that assigns each finite set its number of elements, and each infinite set the value ∞ :

$$\mu(A) \stackrel{\text{def}}{=} \begin{cases} |A| & \text{if } A \text{ is finite} \\ \infty & \text{if } A \text{ is infinite} \end{cases}$$

Example 0.17 (Counting measure on the non-negative integers). For the counting measure μ on the **non-negative integers** $\{0, 1, 2, \dots\}$, $\mu(\{0, 1, 2\}) = 3$, and the set of even numbers has $\mu(\{0, 2, 4, \dots\}) = \infty$. The even numbers are the union of the pairwise disjoint sets $\{0\}, \{2\}, \{4\}, \dots$, and countable additivity agrees: $\sum_{k=0}^{\infty} \mu(\{2k\}) = 1 + 1 + \dots = \infty$.

Definition 0.13 (Measurable space). A **measurable space** $(S, \mathcal{S})$ is a set S together with a σ -algebra $\mathcal{S}$ on S ; the sets in $\mathcal{S}$ are its **measurable sets**.

Example 0.18 (Measurable spaces of die rolls). For the die rolls $D = \{1, 2, 3, 4, 5, 6\}$:

- $(D, \mathcal{S})$, with $\mathcal{S}$ the collection of all subsets of D , is a measurable space (Example 0.1).
- $(D, \{\emptyset, \{2, 4, 6\}, \{1, 3, 5\}, D\})$ is also a measurable space. Its measurable sets are only those four; for example, $\{1\}$ is not one of them.

Definition 0.14 (Measure space). A **measure space** $(S, \mathcal{S}, \mu)$ is a **measurable space** together with a **measure** μ on $\mathcal{S}$.

Example 0.19 (Measure space of die rolls). For the die rolls $D = \{1, 2, 3, 4, 5, 6\}$, adding the counting function $\mu(A) = |A|$ of Example 0.16 to the **measurable space** $(D, \mathcal{S})$ of Example 0.18 makes $(D, \mathcal{S}, \mu)$ a measure space.

Definition 0.15 (Lebesgue measure). **Lebesgue measure** on $\mathbb{R}$ is the **measure** λ on the **Borel σ -algebra** $\mathcal{B}$ that assigns each **interval** its length: for real numbers $a \leq b$,

$$\begin{aligned}\lambda([a, b]) &= \lambda((a, b)) \\ &= \lambda([a, b)) \\ &= \lambda((a, b]) \\ &\stackrel{\text{def}}{=} b - a,\end{aligned}$$

and $\lambda(I) \stackrel{\text{def}}{=} \infty$ for each unbounded interval I .

Lebesgue measure on an interval I , such as $[0, 1]$, is λ applied only to the Borel sets that are subsets of I .

Remark 0.6 (Lebesgue measure exists). Exactly one measure on $\mathcal{B}$ assigns every interval its length; constructing it takes more measure theory than these notes cover (see [Wikipedia: Lebesgue measure](#), which also extends λ to a larger σ -algebra, the Lebesgue-measurable sets, with the same values on Borel sets).

Example 0.20 (Lengths of some Borel sets).

- $\lambda([2, 5]) = 5 - 2 = 3$.
- A single point has length zero: $\lambda(\{a\}) = \lambda([a, a]) = a - a = 0$.
- The integers are the union of the pairwise disjoint single points $\{0\}, \{1\}, \{-1\}, \{2\}, \{-2\}, \dots$ (Example 0.5), so countable additivity gives $\lambda(\mathbb{Z}) = 0 + 0 + \dots = 0$.
- $\lambda([0, \infty)) = \infty$, because $[0, \infty)$ is unbounded.
- Lebesgue measure on $[0, 1]$ gives the whole interval $[0, 1]$ the value $\lambda([0, 1]) = 1 - 0 = 1$.

Remark 0.7 (Measures generalize size). A **measure** generalizes size: it can measure how many elements a set has, as in Definition 0.12, or how long a set of real numbers is, as Lebesgue measure (Definition 0.15) does, assigning each **interval** $[a, b]$ its length $b - a$.

For example, the counting measure on $\{0, 1, 2, \dots\}$ gives $\{2, 3, 4, 5\}$ the value 4, while the Lebesgue measure of the interval $[2, 5]$ is $5 - 2 = 3$. Both can serve as the measures μ_1 and μ_2 in the [Fubini–Tonelli theorem](#).

Definition 0.16 (Measure zero (null set)). Let $(S, \mathcal{S}, \mu)$ be a [measure space](#). A set $A \in \mathcal{S}$ has **measure zero** (also called a **null set** for μ) if $\mu(A) = 0$.

Example 0.21 (Sets of measure zero).

- For a [counting measure](#), only $\emptyset$ has measure zero: a nonempty set has at least one element, so its counting measure is at least 1.
- For Lebesgue measure, every single point $\{a\}$ and the integers $\mathbb{Z}$ have measure zero (Example 0.20), although neither set is empty.

Definition 0.17 (Almost everywhere (μ -almost everywhere, μ -a.e.)). Let $(S, \mathcal{S}, \mu)$ be a [measure space](#). A statement about the elements of S holds **μ -almost everywhere** (written μ -a.e., or **almost everywhere** when μ is clear from context) if the set of elements for which it fails is a [subset](#) of a set of [measure zero](#).

Example 0.22 (Non-negative except on the integers). Let $g : \mathbb{R} \rightarrow \mathbb{R}$ with $g(x) = -1$ when x is an integer and $g(x) = x^2$ otherwise. The statement “ $g(x) \geq 0$ ” fails exactly for $x \in \mathbb{Z}$, and $\lambda(\mathbb{Z}) = 0$ (Example 0.20), so $g \geq 0$ holds λ -almost everywhere, though not everywhere.

For a counting measure, only $\emptyset$ has measure zero (Example 0.21), so a statement holds almost everywhere only if it holds everywhere.

Definition 0.18 (Finite and σ -finite measures). Let $(S, \mathcal{S}, \mu)$ be a [measure space](#). The measure μ is **finite** if $\mu(S) < \infty$. It is **σ -finite** if S is the union $\bigcup_{i=1}^{\infty} A_i$ of a sequence of sets $A_1, A_2, \dots$ in $\mathcal{S}$ with $\mu(A_i) < \infty$ for every i .

Example 0.23 (Finite and σ -finite measures).

- Every finite measure is σ -finite: take $A_1 = S$ and $A_2 = A_3 = \dots = \emptyset$, with $\mu(A_1) = \mu(S) < \infty$ and $\mu(\emptyset) = 0$.
- The counting measure on the die rolls D is finite, with $\mu(D) = 6$.
- The counting measure on $\{0, 1, 2, \dots\}$ is not finite, but it is σ -finite: it is the union of the sets $A_i = \{0, 1, \dots, i\}$, each with $\mu(A_i) = i + 1 < \infty$.
- Lebesgue measure on $\mathbb{R}$ is not finite, since $\lambda(\mathbb{R}) = \infty$, but it is σ -finite: $\mathbb{R}$ is the

union of the intervals $A_i = [-i, i]$, each with $\lambda(A_i) = 2i < \infty$.

Probability measures

Definition 0.19 (Probability measure). A **probability measure** on a measurable space $(\Omega, \mathcal{F})$ is a **measure** P on $\mathcal{F}$ that gives the whole set Ω the value 1:

$$P(\Omega) = 1$$

Then $(\Omega, \mathcal{F}, P)$ is a **probability space**, Ω is its **sample space**, the elements of Ω are **outcomes**, and the sets in $\mathcal{F}$ are **events**.

Example 0.24 (A fair die, and the unit interval).

- For one roll of a fair die, the sample space is $\Omega = D = \{1, 2, 3, 4, 5, 6\}$, the events are all subsets of D , and $P(A) \stackrel{\text{def}}{=} |A|/6$. P is the counting function $\mu(A) = |A|$ of Example 0.16 divided by 6. So $P(\emptyset) = 0$, and P is countably additive, because each sum in Example 0.12 has at most six nonzero terms, and dividing a finite sum by 6 divides each term by 6. It is a probability measure, since $P(D) = 6/6 = 1$. The outcome 3 is in the event “the roll is odd”, $\{1, 3, 5\}$, which has $P(\{1, 3, 5\}) = 3/6 = 1/2$.
- Lebesgue measure on $[0, 1]$ is a probability measure, because $\lambda([0, 1]) = 1$ (Example 0.20).
- The counting measure on D is not a probability measure, because it gives D the value 6.

Remark 0.8 (Mutually exclusive events). In probability, **pairwise disjoint events** are usually called **mutually exclusive**. For example, for one roll of a six-sided die, the events “the roll is **even**”, $\{2, 4, 6\}$, and “the roll is odd”, $\{1, 3, 5\}$, are mutually exclusive: no roll is both even and odd.

Measurable functions and integrals

Definition 0.20 (Measurable function). Let $(S, \mathcal{S})$ be a measurable space. A function $f : S \rightarrow \mathbb{R}$ is **measurable** (with respect to $\mathcal{S}$) if, for every real number c , the set of elements where f is at most c is measurable:

$$\{s \in S : f(s) \leq c\} \in \mathcal{S}$$

Remark 0.9 (Measurable functions in general). Other sources define a function $f : S \rightarrow T$ between measurable spaces $(S, \mathcal{S})$ and $(T, \mathcal{T})$ to be measurable when $\{s \in S : f(s) \in B\} \in \mathcal{S}$ for every $B \in \mathcal{T}$. For $T = \mathbb{R}$ with the [Borel \$\sigma\$ -algebra](#), that definition is equivalent to Definition 0.20 (see [Wikipedia: Measurable function](#)). For example, with $B = (-\infty, c]$, the general definition requires $\{s \in S : f(s) \leq c\} \in \mathcal{S}$, which is the condition in Definition 0.20.

Example 0.25 (Measurable and not measurable). On the die rolls D , take the σ -algebra $\mathcal{S} = \{\emptyset, \{2, 4, 6\}, \{1, 3, 5\}, D\}$ of Example 0.1.

- Let $f(x) = 1$ for even x and $f(x) = 0$ for odd x . The set $\{x \in D : f(x) \leq c\}$ is $\emptyset$ when $c < 0$, $\{1, 3, 5\}$ when $0 \leq c < 1$, and D when $c \geq 1$. All three are in $\mathcal{S}$, so f is measurable.
- Let $g(x) = x$. Then $\{x \in D : g(x) \leq 1\} = \{1\}$, which is not in $\mathcal{S}$, so g is not measurable.

With the σ -algebra of all subsets of D instead, every set $\{x \in D : g(x) \leq c\}$ is a subset of D , so every function $D \rightarrow \mathbb{R}$, including g , is measurable.

Definition 0.21 (Integral of a non-negative function). Let $(S, \mathcal{S}, \mu)$ be a [measure space](#), and let $f : S \rightarrow [0, \infty)$ be [measurable](#). The **integral** of f with respect to μ is

$$\int_S f d\mu \stackrel{\text{def}}{=} \sup \left\{ \sum_{i=1}^n \left(\inf_{s \in A_i} f(s) \right) \mu(A_i) \right\}$$

where the [supremum](#) is over all ways of writing S as the union of finitely many nonempty [pairwise disjoint](#) sets $A_1, \dots, A_n$ in $\mathcal{S}$, and each $\inf_{s \in A_i} f(s)$ is the [infimum](#) of the values of f on A_i . In the sums, $0 \cdot \infty \stackrel{\text{def}}{=} 0$, and $x \cdot \infty \stackrel{\text{def}}{=} \infty$ for $x > 0$; the supremum is ∞ if some sum is ∞ . The integral is also written $\int_S f(s) d\mu(s)$.

(Billingsley 1995, sec. 15)

Remark 0.10 (Each sum is a lower estimate). Each sum in Definition 0.21 splits S into pieces and weights each piece A_i by the smallest value f approaches on it, so each sum is at most the integral. For Lebesgue measure on a closed interval $[a, b]$ and a function $f \geq 0$ that has a [Riemann integral](#) on $[a, b]$, the two integrals agree: $\int_{[a,b]} f d\lambda = \int_a^b f(x) dx$ (see [Wikipedia: Lebesgue integral](#)).

Example 0.26 (Integrals with respect to counting measure). Let μ be the counting measure on the die rolls D , and let $g : D \rightarrow [0, \infty)$. Every function on D is measurable (Example 0.25).

The integral is the sum of the values. For any split of D into nonempty pairwise disjoint sets $A_1, \dots, A_n$:

$$\begin{aligned} \sum_{i=1}^n \left(\inf_{x \in A_i} g(x) \right) \mu(A_i) &= \sum_{i=1}^n \sum_{x \in A_i} \inf_{y \in A_i} g(y) \quad (\mu(A_i) \text{ counts the elements of } A_i) \\ &\leq \sum_{i=1}^n \sum_{x \in A_i} g(x) \quad (\text{an infimum is at most each value}) \\ &= \sum_{x \in D} g(x) \quad (\text{each } x \in D \text{ is in exactly one } A_i) \end{aligned}$$

The split into the six single points $\{1\}, \dots, \{6\}$ gives exactly $\sum_{x \in D} g(x)$, so the supremum is $\int_D g d\mu = \sum_{x \in D} g(x)$. For example, with $g(x) = x$, $\int_D g d\mu = 1 + 2 + \dots + 6 = 21$.

An infinite integral. For the counting measure μ on $\mathbb{N}$ and the constant function $h(n) = 1$, the split with one set, $A_1 = \mathbb{N}$, gives the sum $1 \cdot \mu(\mathbb{N}) = 1 \cdot \infty = \infty$, so $\int_{\mathbb{N}} h d\mu = \infty$.

Definition 0.22 (Integral of a real-valued function). Let $(S, \mathcal{S}, \mu)$ be a [measure space](#), and let $f : S \rightarrow \mathbb{R}$ be [measurable](#). The **positive part** and **negative part** of f are the non-negative functions

$$\begin{aligned} f^+(s) &\stackrel{\text{def}}{=} \max\{f(s), 0\} \\ f^-(s) &\stackrel{\text{def}}{=} \max\{-f(s), 0\} \end{aligned}$$

so that $f = f^+ - f^-$ and $|f| = f^+ + f^-$. The **integral** of f with respect to μ is

$$\int_S f d\mu \stackrel{\text{def}}{=} \int_S f^+ d\mu - \int_S f^- d\mu,$$

using Definition 0.21 for each term, when at least one of the two terms is finite; otherwise, the integral of f is undefined. (f^+ , f^- , and $|f|$ are measurable when f is (Billingsley 1995, Theorem 13.3).)

(Billingsley 1995, sec. 15)

Example 0.27 (An integral with positive and negative parts). For the counting measure μ on the die rolls D , let $f(x) = x - 3$. Its values at $1, \dots, 6$ are $-2, -1, 0, 1, 2, 3$, so:

- f^+ has values $0, 0, 0, 1, 2, 3$, and $\int_D f^+ d\mu = 6$ by Example 0.26;
- f^- has values $2, 1, 0, 0, 0, 0$, and $\int_D f^- d\mu = 3$;
- $\int_D f d\mu = 6 - 3 = 3$, which is also $\sum_{x \in D} (x - 3) = 21 - 18 = 3$.

Definition 0.23 (Absolutely integrable function (integrable function, absolute integrability)). Let $(S, \mathcal{S}, \mu)$ be a [measure space](#). A [measurable function](#) $f : S \rightarrow \mathbb{R}$ is **absolutely integrable** with respect to μ (often just called **integrable**) if the integral of its [absolute value](#) is finite:

$$\int_S |f| d\mu < \infty$$

The property itself is called **absolute integrability**.

Remark 0.11 (An absolutely integrable function has a finite integral). If f is absolutely integrable, then $\int_S f^+ d\mu$ and $\int_S f^- d\mu$ are both finite, because $f^+ \leq |f|$ and $f^- \leq |f|$, so each sum in Definition 0.21 for f^+ or f^- is at most the matching sum for $|f|$. So $\int_S f d\mu$ (Definition 0.22) is a real number.

Example 0.28 (Absolutely integrable and not).

- For the counting measure on the die rolls D , $f(x) = x - 3$ (Example 0.27) is absolutely integrable: $\int_D |f| d\mu = 2 + 1 + 0 + 1 + 2 + 3 = 9 < \infty$ by Example 0.26.
- For the counting measure on $\mathbb{N}$, the constant function $h(n) = 1$ is not absolutely integrable, because $\int_{\mathbb{N}} |h| d\mu = \int_{\mathbb{N}} h d\mu = \infty$ (Example 0.26).

L^p spaces

Definition 0.24 (L^p space and L^p norm ($L^p(\mu)$, $\mathcal{L}^p$)). Let $(S, \mathcal{S}, \mu)$ be a [measure space](#). For any real number $p \geq 1$, the L^p **norm** of a [measurable function](#) $f : S \rightarrow \mathbb{R}$ is:

$$\|f\|_p \stackrel{\text{def}}{=} \left(\int_S |f|^p d\mu \right)^{1/p}.$$

The space $\mathcal{L}^p(S, \mathcal{S}, \mu)$ consists of all measurable functions $f : S \rightarrow \mathbb{R}$ for which $\|f\|_p < \infty$. To satisfy the definiteness axiom of a [norm](#) ($\|f\|_p = 0$ if and only if $f = 0$), functions that are equal [almost everywhere](#) (μ -a.e.) are identified into equivalence classes. The resulting normed vector space is the L^p **space**, denoted $L^p(S, \mathcal{S}, \mu)$ (or simply $L^p(\mu)$).

For $p = \infty$, the **essential supremum norm** (or L^∞ **norm**) is:

$$\|f\|_\infty \stackrel{\text{def}}{=} \inf\{C \geq 0 : \mu(\{s \in S : |f(s)| > C\}) = 0\}.$$

The space $L^\infty(S, \mathcal{S}, \mu)$ consists of equivalence classes of essentially bounded measurable functions.

Example 0.29 (An L^1 function not in L^2). Consider the interval $(0, 1)$ equipped with the Borel σ -algebra (Definition 0.4) and Lebesgue measure μ (Definition 0.15). Define $f : (0, 1) \rightarrow \mathbb{R}$ by $f(x) = x^{-1/2}$.

1. For $p = 1$:

$$\begin{aligned}\|f\|_1 &= \int_0^1 x^{-1/2} dx \\ &= [2x^{1/2}]_0^1 \\ &= 2 \\ &< \infty,\end{aligned}$$

so $f \in L^1((0, 1), \mu)$.

2. For $p = 2$:

$$\begin{aligned}|f(x)|^2 &= (x^{-1/2})^2 \\ &= \frac{1}{x}.\end{aligned}$$

The integral is:

$$\begin{aligned}\int_0^1 \frac{1}{x} dx &= \lim_{\varepsilon \rightarrow 0^+} [\ln x]_\varepsilon^1 \\ &= \lim_{\varepsilon \rightarrow 0^+} (-\ln \varepsilon) \\ &= \infty.\end{aligned}$$

Therefore, $\|f\|_2 = \infty$, so $f \notin L^2((0, 1), \mu)$.

This shows that $L^2 \subset L^1$ on finite-measure sets: higher exponents impose stricter integrability requirements near singularities.

Theorem 0.5 (Holder's inequality). Let $(S, \mathcal{S}, \mu)$ be a *measure space*, and let $p, q \in [1, \infty]$ be conjugate exponents satisfying $\frac{1}{p} + \frac{1}{q} = 1$ (with the convention that $q = \infty$ when $p = 1$, and $q = 1$ when $p = \infty$). If $f \in L^p(\mu)$ and $g \in L^q(\mu)$ (Definition 0.24), then their product fg is *absolutely integrable* ($fg \in L^1(\mu)$) and satisfies:

$$\|fg\|_1 = \int_S |fg| d\mu \leq \|f\|_p \|g\|_q.$$

When $p = q = 2$, this inequality is the Cauchy-Schwarz inequality for integrals.

Theorem 0.6 (Minkowski's inequality (L^p triangle inequality)). Let $(S, \mathcal{S}, \mu)$ be a *measure space* and let $1 \leq p \leq \infty$. For any functions $f, g \in L^p(\mu)$ (Definition 0.24), their sum $f + g$ is in $L^p(\mu)$ and satisfies:

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p.$$

Together with absolute homogeneity $\|cf\|_p = |c| \|f\|_p$ and definiteness, Minkowski's inequality establishes that $L^p(\mu)$ satisfies all axioms of a **normed vector space**.

Example 0.30 (Applying Holder's inequality to finite sums and expectations).

1. **Finite vectors (Cauchy-Schwarz and taxicab bounds):** Equipping a finite set with the **counting measure**, Holder's inequality for conjugate exponents p, q gives:

$$\sum_{i=1}^n |x_i y_i| \leq \left(\sum_{i=1}^n |x_i|^p \right)^{1/p} \left(\sum_{i=1}^n |y_i|^q \right)^{1/q}.$$

Setting $p = 1, q = \infty$ gives $\sum_{i=1}^n |x_i y_i| \leq \|\tilde{x}\|_1 \|\tilde{y}\|_\infty$. Setting $p = q = 2$ gives the standard vector **Cauchy-Schwarz inequality**.

2. **Covariance bound for random variables:** On a probability space with measure P (Definition 0.19), taking $p = q = 2$ gives $E[XY] \leq \sqrt{E[X^2]} \sqrt{E[Y^2]}$, guaranteeing that the covariance between any two square-integrable random variables is finite.

Remark 0.12 (L^p spaces and norms in probability and machine learning). In probability, statistics, and machine learning, L^p spaces structure the geometry of random variables, estimators, and loss functions:

1. **Random variables, moments, and inclusion of L^p spaces:** When $\mu = P$ is a probability measure (Definition 0.19) with total measure $P(S) = 1$, an L^p function is a random variable X with finite p -th moment:

$$\|X\|_p = (E[|X|^p])^{1/p}.$$

By **Jensen's inequality**, if $1 \leq p \leq q \leq \infty$, then $\|X\|_p \leq \|X\|_q$, so $L^q(P) \subseteq L^p(P)$. For instance, every random variable with finite variance ($X \in L^2$) has finite expectation ($X \in L^1$).

2. **The unique Hilbert space L^2 and conditional expectation:** Among all L^p spaces, only $L^2(\mu)$ is an **inner product space**, equipped with the inner product $\langle f, g \rangle \stackrel{\text{def}}{=} \int_S fg d\mu$ (all other L^p spaces fail the parallelogram law). The completeness of L^2 makes it a Hilbert space. In regression, the conditional expectation $E[Y | X]$ is the orthogonal projection of $Y \in L^2$ onto the closed subspace of L^2 functions measurable with respect to X , minimizing the expected mean squared error $\|Y - g(X)\|_2^2$.

3. **Loss functions, estimation, and robustness:** The choice of p in L^p norms defines loss functions for optimization and regression:

- L^2 **loss (mean squared error)**: penalizes large errors quadratically, yielding the conditional mean; sensitive to outliers.
- L^1 **loss (mean absolute error)**: penalizes errors linearly, yielding the conditional median; robust to heavy tails and outliers.
- L^∞ **loss (Chebyshev criterion)**: minimizes the worst-case maximum error across all instances.

Product measures

Definition 0.25 (Product σ -algebra). Let $(S, \mathcal{S})$ and $(T, \mathcal{T})$ be measurable spaces. A measurable rectangle is a Cartesian product $A \times B$ with $A \in \mathcal{S}$ and $B \in \mathcal{T}$. The product σ -algebra $\mathcal{S} \otimes \mathcal{T}$ is the σ -algebra on $S \times T$ generated by the measurable rectangles.

Example 0.31 (Product σ -algebras).

- Give the die rolls D the σ -algebra $\mathcal{S}$ of all its subsets. Each single pair $\{(a, b)\} = \{a\} \times \{b\}$ in $D \times D$ is a measurable rectangle. Every subset of $D \times D$ is a union of finitely many single pairs, and $\mathcal{S} \otimes \mathcal{S}$ contains finite unions of its sets (Theorem 0.1), as well as $\emptyset$, so $\mathcal{S} \otimes \mathcal{S}$ is the collection of all subsets of $D \times D$.
- The rectangle $[0, 2] \times [0, 3]$ is a measurable rectangle for the Borel σ -algebras, so it is in $\mathcal{B} \otimes \mathcal{B}$.

Definition 0.26 (Product measure). Let $(S, \mathcal{S}, \mu)$ and $(T, \mathcal{T}, \nu)$ be measure spaces with σ -finite measures μ and ν . The product measure $\mu \otimes \nu$ is the measure on the product σ -algebra $\mathcal{S} \otimes \mathcal{T}$ that gives each measurable rectangle the product of the measures of its sides:

$$(\mu \otimes \nu)(A \times B) = \mu(A) \nu(B) \quad \text{for all } A \in \mathcal{S} \text{ and } B \in \mathcal{T},$$

with $0 \cdot \infty = \infty \cdot 0 \stackrel{\text{def}}{=} 0$ and $x \cdot \infty = \infty \cdot x \stackrel{\text{def}}{=} \infty$ for $x > 0$. Exactly one measure has these values (Billingsley 1995, Theorem 18.2).

Example 0.32 (Product measures).

- For the counting measure μ on the die rolls D , $(\mu \otimes \mu)(A \times B) = |A||B|$, the number

of pairs in $A \times B$. For example, the set of pairs in which both rolls are even is $\{2, 4, 6\} \times \{2, 4, 6\}$, with $(\mu \otimes \mu)(\{2, 4, 6\} \times \{2, 4, 6\}) = 3 \cdot 3 = 9$ pairs.

- For Lebesgue measure λ on $\mathbb{R}$, $(\lambda \otimes \lambda)([0, 2] \times [0, 3]) = 2 \cdot 3 = 6$, the area of the rectangle.

Further reading

- Billingsley (1995) develops measure theory and builds probability on it. Its early chapters define σ -algebras, measures, and their basic properties.
- Folland (1999) is a graduate text on measure and integration; its first chapters treat σ -algebras and measures.
- Gut (2013) is a graduate course in probability that opens with a chapter on introductory measure theory.

References

Billingsley, Patrick. 1995. *Probability and Measure*. 3rd ed. Wiley Series in Probability and Mathematical Statistics. Wiley.

Folland, Gerald B. 1999. *Real Analysis: Modern Techniques and Their Applications*. 2nd ed. Wiley. <https://www.wiley.com/en-us/Real+Analysis%3A+Modern+Techniques+and+Their+Applications%2C+2nd+Edition-p-9780471317166>.

Gut, Allan. 2013. *Probability: A Graduate Course*. 2nd ed. Springer Texts in Statistics. Springer. <https://doi.org/10.1007/978-1-4614-4708-5>.